

On the Token Reward Quality of On-Policy Distillation

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Abstract

On-policy knowledge distillation trains the student on its own rollouts while the teacher provides token-level soft targets. A common assumption is that the teacher’s supervision is uniformly useful across all token positions. We show this assumption is wrong: teacher signal quality varies dramatically by position. At early positions, the teacher provides an independent assessment of reasoning strategy (high entropy, 21% token-level disagreement with the student); at later positions, the teacher is conditioned on the student’s prefix and largely confirms the student’s choices (low entropy, 7% disagreement). Training on only the first 100 tokens improves performance by +12.2pp over baseline. Critically, training on the last 100 tokens initially improves (+11.4pp) but then *degrades catastrophically* (62.3%→39.8%), mirroring full-sequence collapse—while prefix training remains completely stable. This pinpoints the harmful signal at late response positions. We propose *positional distillation*: restricting KD loss to the first N response tokens, yielding a $12\times$ wall-clock speedup. Experiments across math reasoning (MATH-500), code generation (HumanEval/MBPP), with multiple teacher scales (1.7B–4B) and model families (Qwen, Gemma), show that positional distillation matches or exceeds full-sequence distillation while eliminating training instability.

1 Introduction

On-policy knowledge distillation (KD) trains the student on its own rollouts while the teacher supplies token-level soft targets [Agarwal et al., 2024, Gu et al., 2024]. From the student’s perspective, this is equivalent to receiving a *dense, token-level reward*: at every position the teacher acts as a reward model, scoring each student-generated token via its conditional distribution [Gu et al., 2024, Lu et al., 2025, Yang et al., 2026]. A natural assumption is that this token reward is uniformly useful across positions—that every token in the student’s trajectory receives an equally valuable signal from the teacher. This paper challenges that assumption.

This token reward is typically understood as the teacher assessing the quality of the student’s answer given the question. But since the reward is computed autoregressively— $p_{\text{teacher}}(y_t \mid x, y_{<t})$, where $y_{<t}$ is the *student’s own prefix*—the interaction between teacher and student naturally decomposes into two regimes. At **early positions**, the prefix is largely determined by the prompt x , so the teacher’s assessment is primarily based on the question itself—an *independent* evaluation. At **later positions**, the teacher is conditioned on hundreds of student-generated tokens; it is no longer evaluating the student based on the question alone, but rather scoring how to continue *this particular* student trajectory—a *dependent* evaluation shaped by the student’s prior choices. These two regimes are fundamentally different in nature. This raises a critical question: *is the teacher’s token reward at later positions still high quality?* We investigate this question and find that the answer is no.

Our analysis reveals several converging lines of evidence. **(1) Signal strength decays with position.** Teacher entropy drops from 1.07 at early positions to 0.14 at late positions; teacher-student agreement rises from 58% to 87%. At later positions, the teacher is effectively rubber-stamping the student’s choices. **(2) Late-position supervision is actively harmful.** Training on the last 100 tokens initially improves accuracy (+11.4pp) but then collapses catastrophically (62.3%→39.8%), mirroring full-sequence degradation. Training on the first 100 tokens (+12.2pp) remains completely stable. This localizes the harmful signal to late positions. **(3) Full-sequence training degrades universally.** Full-seq collapses from 62.4% to 37.8% on MATH-500 and from 39.0% to 32.3% on HumanEval, consistent across all teacher scales and model families.

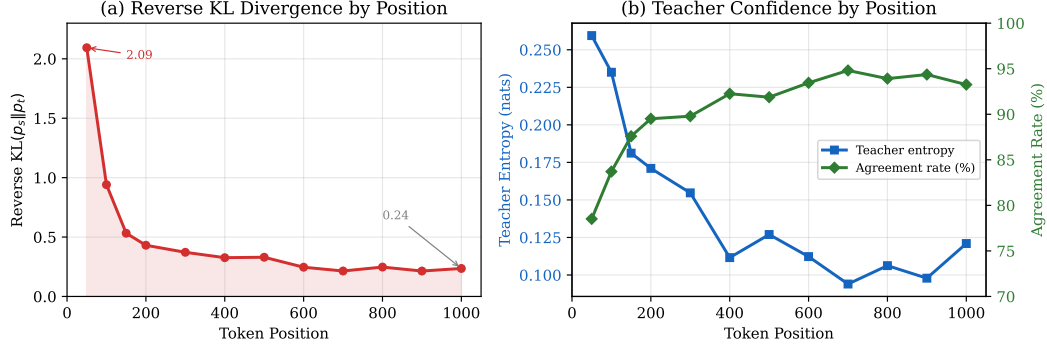


Figure 1: Teacher signal quality degrades with position (100 MATH-500 problems, 1 trajectory each). (a) Reverse KL divergence $KL(p_s || p_t)$ decays from 2.09 in the first 50 tokens to ~ 0.24 by position 1000. Dashed line shows the token-level log-prob gap, which nearly coincides with the reverse KL. (b) Teacher entropy drops from 0.26 to 0.12 while teacher-student agreement rises from 79% to 93%. At later positions, the teacher is effectively rubber-stamping the student’s choices.

These findings motivate **Positional Distillation**: restricting KD loss to the first N response tokens and truncating rollouts accordingly. This is not merely an efficiency trick—it is a principled response to the non-uniform signal quality we identify. Experiments across math reasoning (MATH-500), code generation (HumanEval/MBPP), with teacher scales from 1.7B to 4B and cross-family validation (Qwen, Gemma), show that positional distillation matches or exceeds full-sequence distillation on all tasks while providing $12\times$ wall-clock speedup and eliminating training instability.

Our contributions are:

1. **Analysis of teacher signal quality.** We show that in on-policy KD, the teacher’s supervision quality degrades along the trajectory: early tokens carry strong, independent signal about reasoning strategy, while later tokens contribute weak, dependent signal that is actively harmful when trained on (Section 2).
2. **Positional Distillation.** A method that restricts KD loss to the first N response tokens, simultaneously improving quality, stability, and efficiency (Section 3).
3. **Broad evaluation.** Consistent improvements across math and coding tasks, multiple teacher scales, and two model families (Qwen, Gemma), with detailed ablations on position selection, token selection strategies, loss functions, and training dynamics (Section 4).

2 Token Reward Quality Analysis by Position

We analyze the quality of the teacher’s per-token supervision signal across positions in student-generated trajectories. All analysis in this section uses Qwen2.5-Math-1.5B as the student and Qwen3-1.7B as the teacher on MATH-500, with 59,936 on-policy trajectories (mean length 386 tokens).

2.1 The Conditional Scoring Problem

In on-policy distillation, the teacher scores position t via $p_{\text{teacher}}(\cdot | x, y_{1:t-1}^{\text{student}})$. This score is *not* the teacher’s unconditional opinion about what token should come next—it is the teacher’s opinion *conditioned on the student’s prefix*. As t increases, the prefix $y_{1:t-1}$ becomes increasingly student-determined, and the teacher’s scoring increasingly reflects “how to continue this student trajectory” rather than “what is the right approach.”

This creates a *phase shift* in the teacher’s role along the trajectory. At early positions, the teacher acts as a **strategic guide**—providing independent judgment on how to approach

the problem. At later positions, the teacher degenerates into a **syntax checker**—merely confirming that the student’s execution is locally coherent within the student’s chosen approach. If the student’s strategy is suboptimal, the later-position signal teaches the student to better execute a suboptimal approach—rather than to choose a better one.

2.2 Per-Position KL Distribution

To investigate the distillation signal, we generate 59,936 on-policy trajectories (mean length 386 tokens) from the undistilled student and compute per-position KL divergence against the teacher. Positions are 0-indexed throughout (position 0 is the first generated token).

Figure 1 shows two complementary views of the signal distribution. The reverse KL divergence (left) reveals that the distillation signal is massively front-loaded: the first 50 tokens ($KL = 2.09$) carry $8.9\times$ more signal than the 900–1000 range ($KL = 0.24$), with a steep initial decay followed by a long, slowly declining tail beyond position 200. The teacher confidence analysis (right) explains why: at early positions, the teacher’s distribution has higher entropy (0.26 nats) and lower agreement with the student (79%). By position 1000, entropy drops to 0.12 and agreement rises to 93%—the teacher is no longer providing independent guidance, but merely confirming the student’s choices.

This is a direct consequence of the conditional scoring problem (§2.1): once the student’s prefix determines the reasoning approach, the teacher’s conditional distribution collapses to near-agreement. The residual disagreement at late positions is dominated by formatting conventions (LaTeX delimiters, code patterns) rather than reasoning content.

Strategy vs. execution. Token-type analysis (Appendix D) reveals that early high-KL tokens are predominantly *planning* tokens (“To”, “First”, “Let”)—reflecting strategy disagreement—while late high-KL tokens are formatting conventions. Figure 2 illustrates this with a concrete example: the first 100 tokens establish the solution approach, while the last 100 tokens execute the computation.

2.3 Test-Time Validation: Prefix Continuation

We directly test whether early tokens determine response quality via prefix continuation (Figure 2, left). With a 100-token student prefix, the teacher achieves 62.70% avg@4—only 2.6pp below its baseline. At 300 tokens, performance falls to 51.75%, nearly matching the student baseline: the student’s choices become entrenched. The rapid degradation quantifies the *window of effective influence*: the first ~ 100 tokens are where reasoning strategy is still malleable.

2.4 Training-Time Validation: Position Region Ablation

The analysis above predicts that early-position training should be most effective. We test this by training on different 100-token regions selected from full-length student generations (Figure 2, right).

First-100 (prefix) improves avg@4 by +12.2pp and outperforms all other selection strategies. Middle-100 (+10.9pp) and last-100 (+11.4pp) initially improve over baseline, but their training dynamics diverge sharply: **last-100 degrades catastrophically** from 62.3% to 39.8% at step 150—nearly the same collapse as full-sequence training (62.4%→37.8%). In contrast, middle-100 and prefix remain stable. This reveals that the harmful signal is concentrated specifically at *late* positions (the final $\sim 25\%$ of the response), not uniformly across non-prefix tokens. Information-theoretic selection fares even worse: top-KL-100 achieves only +2.4pp despite capturing 95%+ of total KL. This confirms that **position, not signal strength, is the key criterion**, with early contiguous tokens providing the best combination of quality and stability.

2.5 Summary

Our analysis reveals a coherent picture: (1) the teacher’s signal is strong and independent at early positions, where reasoning strategy is being determined; (2) the signal decays to near-agreement at later positions, where the teacher is conditioned on the student’s prefix;

Problem: In an isosceles right triangle, the altitude to the hypotenuse has length $4\sqrt{2}$. What is the area?

First 100 tokens (strategy): “Let’s start by understanding the problem. In an isosceles right triangle, the two legs are equal, and the hypotenuse is $\sqrt{2}$ times the leg. If we denote each leg by a , then hypotenuse $= a\sqrt{2}$. The altitude to the hypotenuse is half the leg, because the altitude bisects the...”

Last 100 tokens (execution): “...altitude from the right-angle vertex to the hypotenuse. This altitude has length $h = \frac{a \cdot a}{a\sqrt{2}} = \frac{a^2}{a\sqrt{2}} = \frac{a}{\sqrt{2}} = \frac{a\sqrt{2}}{2}$. Setting this equal to $4\sqrt{2}$: $\frac{a\sqrt{2}}{2} = 4\sqrt{2}$, so $a\sqrt{2} = 8\sqrt{2}$, giving $a = 8$. The area is $\frac{1}{2} \times 8 \times 8 = \boxed{32}$.”

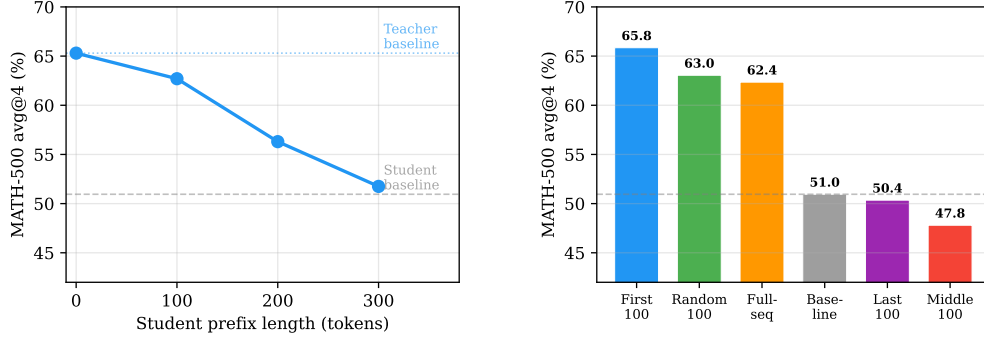


Figure 2: Early tokens encode strategy; later tokens merely execute. *Top:* example showing first 100 tokens (blue) establish the approach while last 100 tokens (orange) execute computation. *Bottom left:* prefix continuation—teacher performance degrades to student baseline by 300 tokens. *Bottom right:* best-checkpoint accuracy per 100-token selection region. First-100 yields +12.2pp over baseline; middle-100 (+10.9pp) and last-100 (+11.4pp) are weaker.

(3) training on early positions captures the most valuable signal, while full-sequence training leads to instability over continued training (detailed in Section 4.3). This motivates restricting distillation to early positions.

3 Method: Positional Distillation

3.1 Formulation

In standard on-policy reverse KL distillation, the student p_θ generates a response $\mathbf{x} = (x_1, \dots, x_T)$ conditioned on a prompt, and the loss is computed against the teacher q_ϕ :

$$\mathcal{L}_{\text{full}} = \mathbb{E}_{\mathbf{x} \sim p_\theta} \left[\sum_{t=1}^T \text{KL}(q_\phi(\cdot | \mathbf{x}_{<t}) \parallel p_\theta(\cdot | \mathbf{x}_{<t})) \right]. \quad (1)$$

Positional distillation restricts the loss to the first N response tokens:

$$\mathcal{L}_{\text{pos}} = \mathbb{E}_{\mathbf{x} \sim p_\theta} \left[\sum_{t=1}^N \text{KL}(q_\phi(\cdot | \mathbf{x}_{<t}) \parallel p_\theta(\cdot | \mathbf{x}_{<t})) \right], \quad (2)$$

where N is a fixed position cutoff ($N \ll T$ in practice). If the student generates an EOS token before position N , loss is computed only up to EOS.

In practice, we both restrict the loss *and* truncate the rollout to N tokens. This couples two benefits: loss masking avoids low-quality tail positions (stability), while rollout truncation reduces generation cost (efficiency). Since autoregressive generation is the dominant cost in on-policy KD, truncating from $T \sim 400$ to $N = 100$ tokens yields 20–35 $\times$ wall-clock speedup (Table 2).

Table 1: Main results. Best performance across training steps for each configuration. **Bold:** best per teacher. Full-seq results use the best step before degradation. Time: seconds per training step (gen + score + train). Gemma uses gemma-2-2b-it student $\rightarrow$ gemma-3-4b-it teacher.

Teacher	Method	MATH-500 avg@4	HumanEval pass@1	MBPP pass@1	Time (s/step)
<i>Student: Qwen2.5-Math-1.5B (LoRA)</i>					
—	Baseline	50.95	32.93	52.12	—
Qwen3-1.7B	Full-seq	62.35	39.02	51.06	~ 133
	Ours ($N=50$)	61.70	37.80	52.12	~ 4
	Ours ($N=100$)	63.15	39.63	51.32	~ 8
Qwen3-4B	Full-seq	65.00 [†]	36.00	51.06	~ 210
	Ours ($N=50$)	61.70	40.24	—	~ 4
	Ours ($N=100$)	64.20	40.85	50.26	~ 8
<i>Student: gemma-2-2b-it (LoRA) — cross-family</i>					
—	Baseline	13.45	23.17	40.48	—
gemma-3-4b-it	Full-seq	11.70 [§]	21.95 [§]	44.44 [§]	~ 210
	Ours ($N=50$)	25.80	31.10	47.35	~ 4
	Ours ($N=100$)	23.15	32.32	51.06	~ 8

[§] Full-seq best step only; degrades severely after (math: 8.35–22%, HE: 22 $\rightarrow$ 11%).

[†] 4B full-seq peaks at step 150 (65.00%), then degrades to 45.75% by step 200.

4 Generalization Experiments

4.1 Setup

Models. We evaluate across multiple student-teacher pairs. Our primary student is Qwen2.5-Math-1.5B [Qwen Team, 2024], paired with teachers of increasing scale: Qwen3-1.7B, Qwen3-4B, and Qwen3-8B [Qwen Team, 2025]. To test cross-family generality, we also evaluate with gemma-2-2b-it (student) and gemma-3-4b-it (teacher), which use different tokenizers requiring cross-tokenizer re-tokenization and shared-vocabulary KL computation.

Training. All experiments use LoRA [Hu et al., 2022] ($r=32$, $\alpha=64$, all linear layers, $\text{lr} = 5 \times 10^{-5}$) with reverse KL divergence loss. Temperature 0.7 for on-policy generation. Each training step processes a batch of 16 problems with 1 rollout per problem ($n_{\text{samples}}=1$, batch size 16). We train for 200 steps on math and function calling, 400 steps on coding, saving checkpoints every 50 steps. Training data: 3,200 math problems, 6,400 coding problems (CodeUltraFeedback), and 3,200 function calling examples (glaiive-function-calling-v2). Our method uses $N=100$ unless otherwise specified.

Evaluation. MATH-500 [Hendrycks et al., 2021, Lightman et al., 2023] with $n=4$ samples at temperature 0.7 (reporting avg@4). HumanEval/HumanEval+ and MBPP/MBPP+ [Chen et al., 2021, Austin et al., 2021, Liu et al., 2024] at temperature 0.0 (reporting pass@1). Function calling (BFCL; Yan et al., 2024) results are in Appendix C.

4.2 Main Results

Table 1 presents the unified main results across tasks, teacher scales, and model families. We report the best performance achieved across all training steps for each configuration.

Results. With the Qwen3-1.7B teacher, $N=100$ achieves the best MATH score (63.15% avg@4, +0.8pp over full-seq) at $12\times$ lower training cost. On HumanEval, $N=100$ (39.63%) slightly outperforms full-seq (39.02%), while MBPP remains comparable across all methods (~ 51 –52%). Critically, positional methods remain stable over training while full-seq degrades catastrophically (Section 4.3). Scaling to the 4B teacher improves quality further: $N=100$

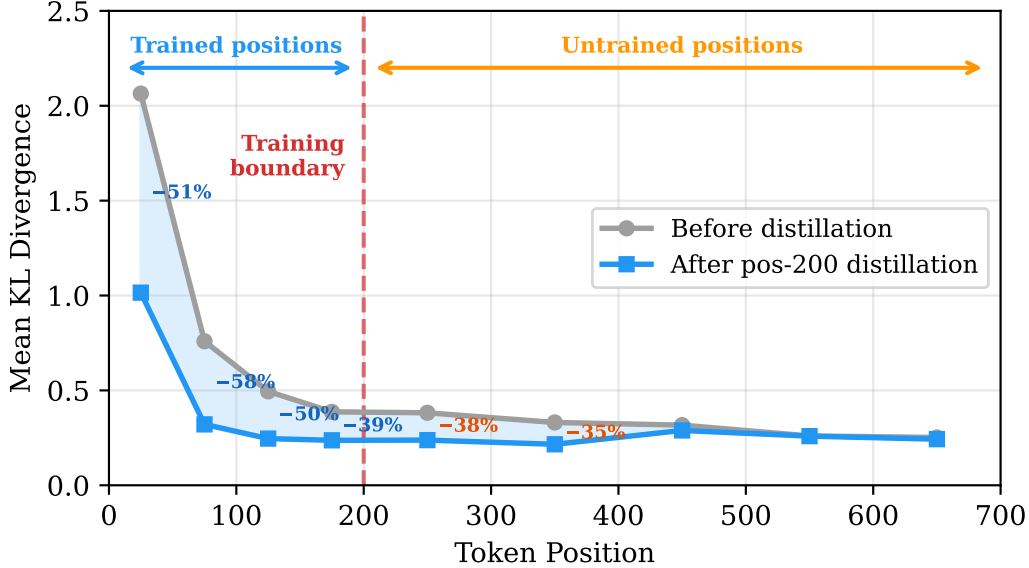


Figure 3: Cascade effect. Pos-200 distillation reduces KL not only in the trained range (0–200) but also in the untrained range (200–400+), confirming that early-token improvements cascade through the autoregressive generation.

achieves 64.20% avg@4 on math and the best HumanEval across both teachers (40.85%). The 4B full-seq peaks at 65.00% (step 150) but then degrades to 45.75% by step 200—the same instability pattern as the 1.7B teacher, only delayed. The Gemma cross-family pair demonstrates that positional distillation transfers across model families with incompatible tokenizers: $N=50$ nearly doubles math accuracy (13.45% $\rightarrow$ 25.80%), while full-seq *degrades below baseline* to 11.70%. Additional teacher scales are in Appendix B.

Cascade effect. A natural concern is that distilling only the first N tokens might only improve those positions. We find the opposite: early-token distillation creates a *cascade effect* that reshapes the entire response (Figure 3). Because the early tokens determine the reasoning strategy, improving them causes the subsequent execution tokens to change accordingly—even though those later positions receive zero distillation signal. Concretely, pos-200 reduces KL by 50–58% within the trained range (positions 0–200) and by 35–39% in the *untrained* range (positions 200–400). This is consistent with the strategy-then-execution structure identified in Section 2.2: better strategy decisions at early positions naturally lead to better execution at later positions.

4.3 Training Stability

Full-sequence distillation degrades with continued training across multiple tasks. On coding, HumanEval drops from 39.0% to 32.3% (–17%) over 200 steps (Figure 4). On math, avg@4 drops from 62.35% (step 100) to 37.75% (step 150)—a catastrophic –24.6pp collapse. The pattern is consistent across teacher scales: the 4B teacher peaks at 65.0% (step 150) then degrades to 45.8% by step 200. In contrast, positional distillation remains stable throughout training on all benchmarks. This is a direct consequence of the signal quality analysis in Section ??: noisy gradients from later positions accumulate over training and distort the student.

4.4 Efficiency

Table 2 compares resource usage. Positional methods (pos-100) require ~ 11 s per step and ~ 9 GB GPU memory, while full-seq requires ~ 133 s per step and ~ 17 GB. This yields a $12\times$

Table 2: Training efficiency comparison. A100-40GB, bs=16. Qwen2.5-Math-1.5B $\rightarrow$ Qwen3-1.7B.

	Ours ($N=100$)	Full-seq
Peak GPU memory	9 GB	17 GB
Time per step	11 s	133 s
Total (200 steps)	37 min	7.4 hours
Speedup	12$\times$	

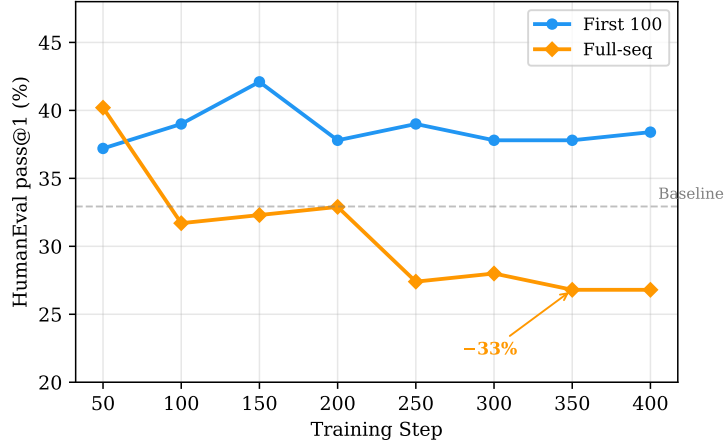


Figure 4: Full-sequence distillation degrades across tasks. On HumanEval, full-seq drops from 39.0% to 32.3% (−17%) over 200 training steps, while positional distillation remains stable. Math shows similar degradation: avg@4 collapses from 62.4% to 37.8% between steps 100–150.

wall-clock speedup. Positional distillation also avoids the need for a vLLM/SGLang inference server, since 50–100 tokens can be generated via standard HF `generate`.

5 Ablation Experiments

5.1 Position Limit Sweep

How sensitive is positional distillation to the choice of N ? Figure 5 shows best avg@4 across $N \in \{5, 10, 20, 50, 100, 150, 200\}$ on MATH-500.

The results reveal a robust sweet spot at $N = 50$ – 200 , where performance saturates. All positional variants from $N \geq 50$ outperform full-sequence distillation (62.35%), with $N=150$ achieving the best result (63.90%) and $N=100$ close behind (63.15%). We use $N=100$ as the default throughout because it offers the best balance of quality and efficiency: 97% of the best performance at 73% of the cost. Even $N=5$ provides +7.3pp over baseline, confirming that the very first tokens carry substantial distillation signal.

5.2 Token Selection Strategy Comparison

A natural alternative to positional selection is *information-theoretic* selection: choose the k tokens with the highest KL divergence or entropy, regardless of position. We compare seven selection strategies, each selecting exactly 100 tokens from full-length student generations (Figure 6).

The results reveal that **prefix** (63.15%) and **ent-teacher** (63.30% peak, but degrades to 61.20%) achieve similar peak performance, followed by **ent-student** (62.70%), **last-100** (62.30%), **middle-100** (61.80%), **random** (61.25%, degrades to 40.70%), and **top-KL** (53.35%). Two patterns emerge. First, the strategy capturing the most signal (top-KL, 95%+ of total KL) performs worst—high-KL tokens are concentrated at later positions where teacher

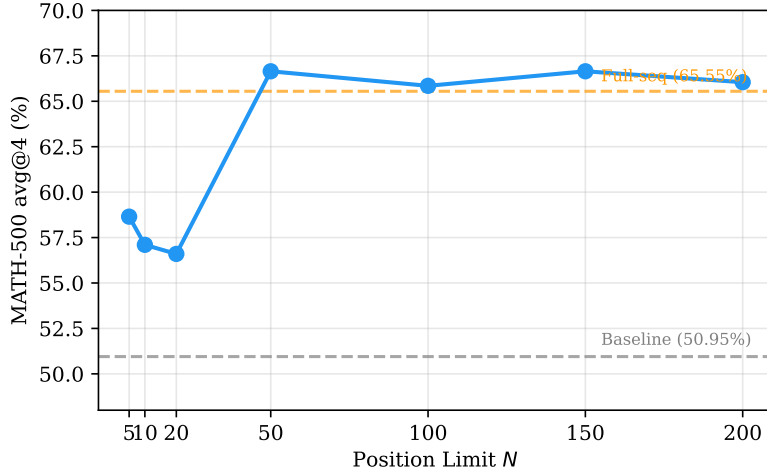


Figure 5: Position limit sweep on MATH-500. Best avg@4 across training steps for each N . Performance increases steadily and surpasses full-seq (62.35%, dashed orange) at $N \geq 50$. Even $N=5$ provides +7.3pp over baseline (50.95%, dashed gray).

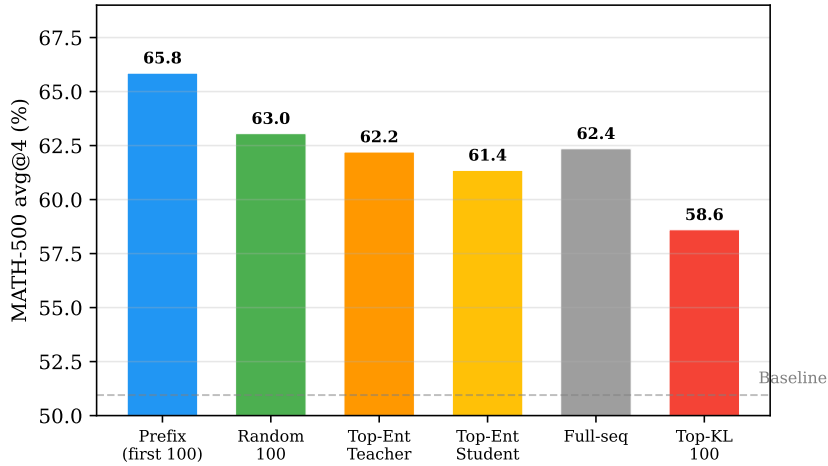


Figure 6: Token selection strategies ($k=100$, best avg@4). Prefix (first 100 tokens) outperforms all alternatives. Middle-100 and last-100 underperform prefix by 1–2pp. Top-KL selects tokens with the highest signal concentration yet performs worst. Random-100 degrades catastrophically.

signal is unreliable. Second, prefix is the only strategy that combines strong peak performance with training stability: ent-teacher matches prefix at peak but degrades by -2.1 pp at step 200, while random collapses entirely. Entropy-based selection partially recovers the prefix advantage because high-entropy tokens are disproportionately early (28–32% in the first 20% of the response; Section 2). This confirms that **contiguous early-position selection provides the best combination of performance and stability**.

Last-100 degrades like full-seq. A striking finding is that last-100 initially performs well (+11.4pp at step 100) but then collapses to 39.8% at step 150—nearly identical to full-sequence degradation (37.8%). Meanwhile, middle-100 remains stable (61.8%→61.8%). This localizes the harmful signal to the *final* positions of the response, where the teacher is most conditioned on the student’s prefix. The last-100 result provides the strongest evidence for our signal quality hypothesis: even selecting the same number of tokens ($k=100$), training

on late positions is unstable while training on early or middle positions is safe. Prefix selection combines the best signal quality (early positions) with complete training stability.

LoRA vs. full fine-tuning. LoRA ($r=32$) substantially outperforms full fine-tuning on both math (+10.0pp avg@4: 63.15% vs. 53.15%) and coding (+4.3pp HumanEval) for pos-100, suggesting that LoRA’s low-rank constraint acts as beneficial regularization. Full results are in Appendix C.

Additional student-teacher pairs. We evaluate with Qwen3-1.7B and Qwen3-4B as students (with 4B and 8B teachers). Distillation is most effective when the student has significant room for improvement: Qwen3-1.7B gains +13.9pp on BFCL from a 4B teacher. Stronger students show diminishing returns (Appendix B).

6 Related Work

Knowledge Distillation for Language Models. Knowledge distillation [Hinton et al., 2015] transfers knowledge from a teacher model to a smaller student by training on soft targets. Kim and Rush [2016] extended this to sequence models with word-level and sequence-level distillation objectives. For large language models, the choice of divergence measure and data distribution has proven critical. Gu et al. [2024] demonstrated that reverse KL divergence is preferable for generative LLM distillation, as it avoids the student overestimating low-probability teacher modes. Agarwal et al. [2024] introduced Generalized Knowledge Distillation (GKD), showing that on-policy generation—where the student generates sequences and the teacher provides soft labels—substantially outperforms supervised (off-policy) distillation. Ko et al. [2024] proposed skew KL divergence and adaptive off-policy approaches for more efficient distillation, while Wen et al. [2023] generalized sequence-level KD to arbitrary f-divergences. Wu et al. [2025] challenged the conventional mode-seeking vs. mean-seeking interpretation of reverse and forward KL in the LLM setting, finding that both divergences converge given sufficient training but focus on different distribution regions in early epochs. More recently, Xu et al. [2025] introduced Speculative Knowledge Distillation, which uses interleaved teacher-student sampling inspired by speculative decoding, filtering tokens the teacher is unlikely to produce. Our work builds on the on-policy reverse KL paradigm of Agarwal et al. [2024] and Gu et al. [2024], but introduces position-based loss computation that fundamentally changes which tokens receive gradient signal.

Token-Level Importance in Distillation and Reasoning. A growing body of work demonstrates that not all tokens contribute equally to learning. Wang et al. [2025] showed that only $\sim 20\%$ of chain-of-thought tokens have high entropy, and these “forking tokens” steer reasoning paths—applying RL gradients to only these tokens matches full-gradient performance. In the RL fine-tuning setting, Vassoyan et al. [2025] demonstrated that uniform KL penalties across all token positions prevent exploration on “critical tokens” where the pretrained model is most uncertain, and proposed entropy-weighted KL relaxation that allows the model to deviate from its prior precisely where it needs to learn. Li et al. [2025] discovered a “preplan-and-anchor” mechanism in LLM attention patterns, where models first perform long-range contextual reference before producing semantic anchor tokens, enabling targeted credit assignment. Singh and Hakkani-Tür [2026] proposed greedy pruning of reasoning chains and showed that LLMs internally encode token-level functional importance for answer generation.

In the distillation setting specifically, several concurrent works have explored token selection. Huang et al. [2025] proposed SelecTKD, a propose-and-verify framework where the teacher verifies student-proposed tokens, applying full loss to accepted tokens and masking rejected ones. Xie et al. [2026] introduced AdaKD, which adaptively adjusts token-level temperature based on learning stability. Tavor et al. [2026] systematically disentangled selective KD along position, class, and sample axes, proposing student-entropy-guided position selection (SE-KD)

that reduces wall time by 70% without sacrificing accuracy. Kim and Baek [2026] combined entropy-based token selection with preference ranking in a dual distillation framework.

These works collectively establish that not all tokens deserve equal treatment during training. Our approach differs in using *absolute position* as the sole selection criterion, requiring no per-token computation of entropy, difficulty, or verification scores. This simplicity is motivated by our empirical finding that KL divergence between teacher and student concentrates at early positions corresponding to reasoning strategy selection. Furthermore, unlike prior token-selection methods that aim primarily at efficiency or robustness, we show that positional loss also eliminates training instability—a qualitative benefit that token-level adaptive methods do not address.

Training Stability and Sequence-Level Issues. Full-sequence on-policy distillation is known to suffer from instability. Holtzman et al. [2020] identified that likelihood-based training can produce bland, repetitive text—a form of neural text degeneration. Shumailov et al. [2024] demonstrated that iteratively training generative models on their own outputs leads to model collapse, where distribution tails disappear. In on-policy KD specifically, Jang et al. [2026] proposed Veto, a geometric bridge in logit space that suppresses harmful gradients on low-confidence tokens, directly addressing the instability we also observe. Our approach is complementary: rather than modifying the target distribution, we restrict the loss scope to early positions where teacher-student agreement is naturally highest, achieving stability through a simpler mechanism.

Several works have studied how sequence length affects distillation quality. Chen et al. [2025] showed that training on only the first 50% of tokens in supervised reasoning distillation retains ~91% of full-sequence performance, establishing a computation-quality tradeoff. Yan et al. [2026] addressed exposure bias and capacity misalignment in long chain-of-thought distillation through temperature-scheduled learning. Our work extends the truncation insight to the on-policy setting: we demonstrate that positional loss masking on student-generated sequences achieves equal or better performance than full-sequence loss, while providing the additional benefit of eliminating degenerate repetition patterns that are unique to on-policy training.

Curriculum Learning and Sequence Scheduling. Curriculum learning [Bengio et al., 2009] improves training by gradually increasing task difficulty. Bengio et al. [2015] applied this principle to sequence models through scheduled sampling, transitioning from teacher-forced to free-running generation during training. In the LLM pretraining setting, Li et al. [2022] showed that long sequences at training start cause extreme gradient variance, and sequence length warmup stabilizes training while enabling larger batch sizes. Pouransari et al. [2024] extended this with variable sequence length curriculum for LLM pretraining, achieving up to 3× faster convergence. Our progressive position schedule—linearly increasing the position limit from 1 to 200 over training—applies the curriculum principle to the distillation loss scope rather than to the input sequence length, and represents a novel form of curriculum specific to knowledge distillation.

Reasoning Distillation. The distillation-vs-RL paradigm for reasoning has been established by DeepSeek-R1 [DeepSeek-AI, 2025], which showed that distilled models can outperform RL-trained ones in certain settings. Chain-of-thought prompting [Wei et al., 2022] enables LLMs to solve complex problems via intermediate reasoning, and distilling CoT capabilities has become a key application of KD. Our work contributes to this landscape by demonstrating that the effectiveness of distillation can be significantly improved through principled control of the loss computation scope.

7 Discussion

The conditional scoring problem is fundamental. Our analysis reveals that a core issue in on-policy distillation is often overlooked: the teacher’s supervision at position t is

not an unconditional assessment of token quality, but a *conditional* assessment given the student’s prefix $y_{<t}$. At early positions, this conditioning is minimal (the prefix is mostly the prompt), so the teacher provides genuinely independent guidance on reasoning strategy. At later positions, the teacher is locked into the student’s chosen approach and can only score execution details within that approach. This “conditional scoring problem” explains both why early-position signal is valuable (it transfers strategy) and why later-position signal is harmful (it reinforces potentially suboptimal approaches).

Broader implications. The signal quality degradation we identify may extend beyond distillation to other on-policy optimization methods that use token-level rewards, including RLHF and DPO. Whenever a reward model or reference model scores tokens conditioned on a learner-generated prefix, the same conditional dependency applies: later-position rewards are increasingly shaped by the learner’s choices rather than representing independent quality assessments. Investigating positional signal quality in these settings is a promising direction.

Limitations. Full-sequence distillation computes loss on $\sim 4\times$ more tokens per step than pos-100, resulting in larger per-step gradient magnitude. We cannot fully disentangle whether full-seq degradation stems from low-quality late-position signal or from the higher effective learning rate; a loss-normalization ablation (dividing by sequence length) would isolate this. All results are single-run without confidence intervals; the small margins between some methods (e.g., prefix vs. last-100) may not be statistically significant. Our analysis focuses on the Qwen model family and reverse KL divergence; extending to other families, larger scales, and forward KL would strengthen generality. The optimal N varies by task; adaptive per-example N selection could be more robust.

Conclusion. We show that in on-policy knowledge distillation, the teacher’s per-token supervision quality is fundamentally non-uniform: early tokens carry strong, independent signal about reasoning strategy, while later tokens provide weak, dependent signal that degrades training stability. This insight motivates Positional Distillation—restricting KD loss to the first N response tokens—which matches or exceeds full-sequence distillation across math and coding tasks while eliminating training instability and achieving $12\times$ speedup. Teaching the student how to begin is sufficient to improve the entire response.

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A Training Efficiency

Table 3: Per-step time breakdown. Wall-clock time on a single A6000 (48GB). Student: Qwen2.5-Math-1.5B, LoRA, bs=16.

Teacher	Method	Gen (s)	Score (s)	Train (s)	Total (s)
Q3-1.7B	Full-seq	100–731	3–10	5–12	~280
	Ours ($N=100$)	~5	~1	~2	~8
Q3-4B	Full-seq	97–733	3–9	6–10	~210
	Ours ($N=100$)	~5	~1	~2	~8
Q3-8B	Full-seq	96–230	7–11	1–6	~170 [†]
	Ours ($N=100$)	~5	~1	~2	~8

[†] Frequent OOMs; 48GB insufficient for 8B teacher + vLLM + student.

B Additional Student-Teacher Pairs

Table 4: Additional student-teacher pairs. All use pos-100, LoRA, 3,200 problems.

Student	Teacher	MATH-500 avg@4	BFCL full_acc	HumanEval pass@1
Qwen3-1.7B	— (baseline)	69.20%	59.80%	39.63%
Qwen3-1.7B	Qwen3-4B	69.20%	73.70%	44.51%
Qwen3-1.7B	Qwen3-8B	69.15%	74.00%	40.24%
Qwen3-4B	— (baseline)	77.95%	73.00%	73.17%
Qwen3-4B	Qwen3-8B	77.05%	76.80%	71.34%

C Full Experimental Results

C.1 Math Results: Per-Step Performance

Table 5 presents per-step results for the primary math experiments (LoRA, $n=1$, 3,200 problems).

Table 5: MATH-500 per-step results (V1 verified). LoRA, $n=1$, bs=16, 3,200 problems. Baseline: 50.95% avg@4.

Method	Step 50	Step 100	Step 150	Step 200
Pos-5	56.90	57.80	58.20	57.00
Pos-10	58.75	54.60	52.40	52.40
Pos-20	57.65	53.70	54.05	55.65
Pos-50	58.05	60.65	61.70	60.85
Pos-100	60.45	62.50	63.15	62.35
Pos-150	62.00	62.95	62.55	63.90
Pos-200	61.00	62.45	62.40	61.30
Full-seq	61.30	62.35	37.75 [†]	—

[†] Catastrophic degradation.

Table 6: Complete MATH-500 results, LoRA, $n=16$, $bs=16$. Best per configuration in bold.

Config	Step 50	Step 100	Step 150	Step 200
<i>avg@4</i>				
Pos-5	55.15	55.65	56.50	56.30
Pos-10	59.50	56.40	56.30	56.00
Pos-20	60.20	57.35	56.40	56.55
Pos-50	62.45	62.15	60.50	61.90
Pos-100	61.15	63.55	64.25	63.85
Pos-150	63.55	65.70	64.80	64.40
Pos-200	63.25	66.75	64.55	65.25
Progressive	61.10	62.05	61.95	62.30
Full-seq [†]	65.60	46.30	49.70	43.80
Full-seq (first-boxed)	65.55	64.95	65.55	64.95
<i>pass@4</i>				
Pos-5	74.0	74.8	76.0	73.4
Pos-10	75.8	73.2	75.4	74.2
Pos-20	77.0	76.0	73.4	75.8
Pos-50	77.0	78.4	77.6	78.6
Pos-100	77.6	79.2	80.2	80.2
Pos-150	79.6	77.4	80.0	79.4
Pos-200	79.6	80.2	79.4	79.2
Progressive	77.2	77.8	78.2	78.8
Full-seq	80.0	72.2	77.0	74.2

[†] Last-boxed extraction; degrades due to repetition.

Table 7: Complete coding results, LoRA. HumanEval (HE) and MBPP pass@1.

Config	Metric	s50	s100	s150	s200	s250	s300	s350	s400
Pos-50	HE	37.8	39.0	39.6	41.5	40.2	40.9	42.1	40.9
	MBPP	52.1	48.9	46.0	46.0	47.6	46.3	46.6	46.3
Pos-100	HE	37.2	39.0	42.1	37.8	39.0	37.8	37.8	38.4
	MBPP	52.1	49.7	49.2	49.2	48.4	49.2	47.9	48.9
Pos-150	HE	36.6	35.4	36.6	39.0	41.5	39.6	38.4	37.2
	MBPP	51.3	50.0	48.4	50.5	49.5	50.3	49.7	48.7
Full-seq	HE	40.2	31.7	32.3	32.9	27.4	28.0	26.8	26.8
	MBPP	52.6	48.9	49.5	48.1	47.9	47.1	48.4	47.6

C.2 Math Results: Earlier Series ($n=16$, 200 problems)

C.3 Coding Results

C.4 Full Fine-Tuning Results

Table 8: MATH-500 per-step results, full fine-tuning (V1 verified). $n=1$, $bs=16$, 3,200 problems, $lr=5\times 10^{-6}$. Baseline: 50.95% avg@4.

Method	Step 50	Step 100	Step 150	Step 200
FullFT Pos-50	51.80	52.20	51.60	53.70
FullFT Pos-100	50.85	52.35	53.00	53.15
FullFT Full-seq	52.70	54.10	54.35	53.50
LoRA Pos-100	60.45	62.50	63.15	62.35

LoRA outperforms FullFT by +10pp, suggesting low-rank regularization is beneficial.

Table 9: Coding results, full fine-tuning. HumanEval (HE) and MBPP pass@1. $n=1$, $bs=16$, 6,400 problems, $lr=5\times 10^{-6}$.

Config	Metric	s50	s100	s150	s200	s250	s300	s350	s400
Pos-50	HE	31.7	35.4	36.0	36.0	36.0	35.4	36.0	34.8
	MBPP	52.6	53.4	52.9	52.4	52.4	52.6	52.9	53.4
Pos-100	HE	31.7	34.1	35.4	34.8	36.6	36.0	36.0	36.6
	MBPP	52.6	54.0	52.6	52.1	53.2	53.4	53.4	52.9
Pos-150	HE	31.7	36.0	36.0	35.4	36.0	36.6	35.4	37.2
	MBPP	53.2	51.9	52.4	52.4	53.2	54.2	53.2	52.4
Pos-250	HE	32.9	34.8	35.4	35.4	37.2	36.6	37.8	33.5
	MBPP	54.0	53.7	54.0	54.8	54.5	53.4	54.2	53.2
Pos-200tok	HE	32.3	36.6	36.6	35.4	36.0	34.8	36.0	35.4
	MBPP	52.1	52.6	54.0	52.6	53.7	54.5	53.2	53.7
Full-seq	HE	32.3	32.3	36.6	36.0	31.1	30.5	31.7	31.1
	MBPP	53.2	53.4	53.2	54.0	51.9	52.9	52.9	53.7

C.5 Token Selection Strategy Results

Table 10: Token selection strategy comparison (V1 verified). All methods generate full-length responses and select $k=100$ tokens for loss computation. MATH-500 avg@4. Baseline: 50.95%.

Strategy	Step 50	Step 100	Step 150	Step 200	Best
Prefix (first 100)	60.45	62.50	63.15	62.35	63.15
Ent-teacher	59.90	62.45	63.30	61.20	63.30 [†]
Ent-student	60.70	62.70	62.35	62.55	62.70
Last-100	58.85	62.30	39.75 [†]	—	62.30
Middle-100	58.90	61.80	61.75	—	61.80*
Random	59.35	61.25	40.70 [†]	40.70 [†]	61.25
Top-KL	43.25	50.05	52.60	53.35	53.35

[†] Degrades with continued training. * Training in progress.

C.6 Loss Function and Weighting Ablations

The results show that (1) full-vocabulary reverse KL outperforms Top-K renormalized KL by +1.25pp on pos-100, confirming that the long tail of the vocabulary distribution carries useful gradient signal; (2) all soft weighting approaches substantially underperform hard positional

Table 11: Loss function ablations (V1 verified). Soft weighting methods use full-seq generation with position-dependent or entropy-dependent per-token weights. Top-K uses renormalized KL over teacher’s top-32 tokens. MATH-500 avg@4.

Method	Step 50	Step 100	Step 150	Step 200	Best
<i>Full-vocab reverse KL (ours)</i>					
Hard pos-100	60.45	62.50	63.15	62.35	63.15
Full-seq	61.30	62.35	37.75 [†]	—	62.35
<i>Top-K renormalized KL (K=32)</i>					
Top-K pos-100	56.90	61.90	61.70	61.75	61.90
Top-K full-seq	61.10	62.20	61.75	61.90	62.20
<i>Soft weighting (full-seq generation)</i>					
Exp decay ($K=100, \lambda=0.015$)	55.40	56.70	54.45	55.80	56.70
Ent-teacher ($\alpha=2.0$)	54.30	57.90	54.10 [†]	—	57.90
Ent-student ($\alpha=2.0$)	55.15	—	—	—	—*
Ent-joint ($\alpha=1.5$)	55.60	—	—	—	—*

[†] Degrades. * Training in progress. Full-vocab KL outperforms Top-K by +1.25pp.

cutoff, indicating that smooth position-dependent downweighting is insufficient—the harmful late-position signal must be completely excluded.

C.7 Function Calling Results

Table 12: Function calling results (BFCL), LoRA. Name accuracy / Full accuracy / Parse rate. Best full_acc in **bold**.

Method	Best Step	Name Acc	Full Acc	Parse Rate
Baseline	—	9.70%	2.70%	24.20%
Teacher (Qwen3-1.7B)	—	75.30%	54.00%	75.30%
Pos-50	200	95.20%	57.20%	98.30%
Pos-100	100	86.20%	61.30%	91.30%
Pos-150	200	88.70%	61.50%	92.50%
Pos-200	200	80.80%	54.50%	90.20%
Full-seq	100	81.00%	58.20%	86.70%

C.8 Scaling Experiment Results

Table 13: Scaling results. All use pos-100, LoRA, 3,200 problems. Best across training steps. Baseline performance shown for reference.

Student → Teacher	Math avg@4	BFCL full_acc	HE pass@1	MBPP+ pass@1
M-1.5B baseline	50.95%	2.70%	—	—
M-1.5B → Q3-1.7B	65.85%	61.30%	42.07%	44.44%
M-1.5B → Q3-4B	68.95%	45.80%	43.29%	47.62%
M-1.5B → Q3-8B	67.85%	57.00%	41.46%	44.44%
Q3-1.7B baseline	69.20%	59.80%	39.63%	44.71%
Q3-1.7B → Q3-4B	69.20%	73.70%	44.51%	47.62%
Q3-1.7B → Q3-8B	69.15%	74.00%	40.24%	48.15%
Q3-4B baseline	77.95%	73.00%	73.17%	55.56%
Q3-4B → Q3-8B	77.05%	76.80%	71.34%	56.35%

C.9 Full-Sequence Degradation Details

Table 14: Boxed repetition progression in full-sequence LoRA distillation on MATH-500 ($n=16$, $bs=16$).

Statistic	Step 50	Step 100	Step 150	Step 200
Multi-boxed responses	2.9%	91.5%	80.5%	89.1%
Avg <code>\boxed{}</code> per response	1.0	88.1	54.1	58.0
Repetitive responses	5.9%	78.8%	58.3%	77.5%
Avg response length (chars)	1,388	4,023	3,585	4,558
Official avg@4	65.60%	46.30%	49.70%	43.80%
Relaxed avg@4 (GT in text)	74.60%	70.90%	71.90%	70.30%

D Token Classification Methodology

We classify each token into six categories based on string matching:

1. **planning**: Reasoning keywords (“To”, “Let”, “First”, “Step”, “We”, “Given”, “Therefore”, “Thus”, “Since”).
2. **structural**: Punctuation, whitespace, formatting tokens.
3. **math_number**: Digits (0–9).
4. **math_operator**: Arithmetic operators (+, −, ×, /, =).
5. **math_latex**: LaTeX delimiters ($\backslash$ (, $\backslash$ [).
6. **continuation**: All others.

Table 15: Mean KL by token category and position range.

Category	0–4	5–19	20–49	50–99	100–199	200–499
planning	4.50	0.79	1.49	1.66	1.49	2.37
structural	3.26	1.46	1.60	0.93	0.60	0.86
math_number	1.49	0.60	0.74	0.28	0.17	0.13
math_operator	7.30	1.84	0.81	0.37	0.21	0.14
math_latex	8.84	9.23	6.50	4.95	2.97	1.87
continuation	1.94	1.12	1.19	0.89	0.70	0.48

Table 16: Top 20 highest-KL tokens (minimum 50 occurrences across 10,000 trajectories).

Rank	Token	Category	Count	Mean KL
1	“Solution”	planning	152	21.93
2	“Analysis”	continuation	125	16.49
3	\[	math_latex	7,152	13.21
4	“examines”	continuation	74	11.51
5	“He”	continuation	150	10.80
6	\(	math_latex	21,243	10.30
7	“First”	planning	1,706	9.98
8	“tests”	continuation	52	8.71
9	\	math_latex	82	8.68
10	“There”	continuation	201	8.28
11	“Therefore”	planning	4,913	7.95
12	“Identify”	continuation	1,345	7.78
13	“To”	planning	8,806	6.90
14	“When”	continuation	89	6.62
15	“This”	continuation	621	6.34
16	“Thus”	planning	2,547	6.24
17	“First” (space)	planning	259	6.10
18	“To” (space)	planning	1,174	5.37
19	“The”	planning	2,626	5.25
20	“Next”	planning	1,753	5.03